

32-bit microcontrollers

HC32 series RTC compensation function and
real-time clock compensation method

Application note

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Applicable objects

Product series	Product model	Product series	Product model	Product series	Product model
HC32L110	ALL	HC32L17x	ALL	HC32L19x	ALL
HC32L07x	ALL	HC32L16x	ALL	HC32L18x	ALL
HC32L13x	ALL	HC32F052	ALL	HC32F472	ALL
HC32F460	ALL	HC32F4A0	ALL	HC32F448	ALL
HC32F45x	ALL	HC32A460	ALL	HC32A4A0	ALL
HC32A472	ALL	HC32A448	ALL	HC32M441	ALL
HC32L021	ALL	HC32A052	ALL	-	-

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1 Overview

Applications with real-time clocks usually have an external 32768Hz high-precision ppm quartz crystal on the hardware board, and its temperature characteristic curve within a certain temperature range is approximately parabolic. In applications, it is necessary to calculate the ppm deviation of the crystal based on the current temperature, and compensate it through the RTC module inside the microprocessor to achieve the correction of the actual output ppm.

This application note mainly includes:

- Compensation function of Xiaohua microprocessor RTC module
- Crystal temperature characteristic curve
- Real-time clock compensation method

Note:

– This application note is a supplementary material for the application of Xiaohua Semiconductor MCU and cannot replace the reference manual. For specific functions and register operations and other related matters, please refer to the reference manual.

2 Introduction to **RTC compensation** function

This chapter introduces the compensation function of Xiaohua microprocessor RTC module.

There are two methods for the compensation function of Xiaohua microprocessor RTC module:

- Error compensation based on its own clock, that is, average compensation within a certain time (such as 32 seconds), in some microprocessor user manuals, it is also called distributed compensation
- Error compensation based on high-speed clock, that is, compensation per second, in some microprocessor user manuals It is also called uniform compensation

This application manual introduces and uses the first method, that is, based on its own clock, average compensation within a certain time.

2.1 **RTC compensation principle**

When the counter selects 32768Hz clock counting, if the accuracy of each second needs to be compensated, it can only be compensated according to the integer cycle of 32768Hz. The minimum unit of compensation per second is $(1 / 32768) * 10^6 = 30.5\text{ppm}$, which cannot meet the high-precision requirements.

The RTC module involved in this application note adopts the method of expanding the compensation cycle to 2^n seconds, so the compensation accuracy per second is calculated as follows:

$$\text{Compensation accuracy} = (1 / 32768) / 2^n * 10^6 \text{ ppm}$$

Formula 1 Compensation accuracy

- When $n=5$, the compensation cycle is expanded to 32 seconds, and the compensation accuracy per second is $(1 / 32768) / 2^5 * 10^6 = 0.954\text{ppm}$

2.2 **Implementation of RTC compensation function**

The RTC compensation function is implemented by introducing a decimal setting method in the register.

For example: when $n=5$, the compensation register is 9 bits, the upper 4 bits are the integer part, and the lower 5 bits are the decimal part.

8	7	6	5	4	3	2	1	0
Integer part				Fractional part				

The integer part refers to the value that needs to be compensated every second in 32 seconds, and the decimal part refers to how many seconds in 32 seconds need to be compensated. For example, when the current clock compensation target value is 185.11ppm:

$$185.11\text{ppm} / (1 / 32768 * 10^6 \text{ ppm}) = 6.0657$$

- Integer part = 6, that is, 6 seconds need to be compensated every second in 32 seconds
- Decimal part = $0.0657 * (1 / 32768 * 10^6) = 2$, that is, 2 seconds in 32 seconds need to be compensated

According to the above description, the maximum compensation value of the 9-bit compensation register with a 5-digit decimal part in a 32-second period is calculated as follows:

$$\text{Maximum compensation value} = (2^9 - 1) * (1 / 32768 / 2^5) * 10^6 = 487.33\text{ppm}$$

Formula 2 Maximum compensation value

But the actual clock ppm deviation value may be positive or negative, because:

$$\text{ppm} = [(\text{actual frequency} - \text{set frequency}) / \text{set frequency}] * 10^6$$

Formula 3 ppm definition

- When the actual frequency > set frequency, that is, the ppm value is positive, the clock cycle needs to be increased, that is, the compensation value is positive
- When the actual frequency < set frequency, that is, the ppm value is negative, the clock cycle needs to be reduced, that is, the compensation value is negative

So the compensation value implemented by the actual register also needs to be set to meet the requirements of positive and negative ppm compensation.

The following table describes the compensation register of HC32L196 when the counting clock is 32768Hz:

Clock error compensation register (RTC_COMPEN)

Reset value: 0x0000 0020

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Res															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
EN	Res						CR								

Bit	Mark	Name	Function description	Access
b31~b16	Res	Reserved	Keep reset value	
b15	EN	Compensation Enable	0: Disable clock error compensation 1: Enable clock error compensation	RW
b14~b9	Res	Reserved	Keep reset value	

By setting the compensation value, an accuracy compensation of +/-0.954ppm can be performed per second.

The compensable range is -274.7ppm ~ 212.7ppm.

Minimum differential error +/-0.477ppm. Minimum resolution 0.954ppm.

For specific compensation accuracy, please refer to the table below:

			Compensation value setting										Compensation number		
			EN	CR[8:0]											
b8~b0	CR	Compensation value	1	1	0	0	0	0	0	0	0	0	-274.7ppm	RW	
				1	0	0	0	0	0	0	0	1	-273.7ppm		
				:	:	:	:	:	:	:	:	:	:		
				0	0	0	0	1	1	1	1	1	-0.954ppm		
				0	0	0	1	0	0	0	0	0	0ppm		
				:	:	:	:	:	:	:	:	:	:		
				0	1	1	1	1	1	1	1	1	+211.7ppm		
				0	1	1	1	1	1	1	1	1	+212.7ppm		
			0	X	X	X	X	X	X	X	X	X	No compensation		

- When the target value to be compensated = $(32767.71 - 32768) / 32768 * 10^6 = -8.85\text{ppm}$, the compensation step size = $-8.85 / (1 / 32768 / 2^5 * 10^6) = -9.28 \approx -9$

Because the register value of 0ppm is 0x20, the value written to the register = $0x20 - 9 = 23 = 0b000010111$

- When the target value to be compensated = $(32766.93 - 32768) / 32768 * 10^6 = -32.65\text{ppm}$, the compensation step size = $-32.65 / (1 / 32768 / 2^5 * 10^6) = -34.24 \approx -34$

Because the register value of 0ppm is 0x20, so the value written to the register = $0x200 + (0x20 - 34) = 510 = 0b111111110$

- When the target value to be compensated = $(32768.91 - 32768) / 32768 * 10^6 = 27.775\text{ppm}$, the compensation step = $27.775 / (1 / 32768 / 2^5 * 10^6) = 29.12 \approx 29$

Because the register value of 0ppm is 0x20, the value written to the register = $0x20 + 29 = 61 = 0b000111101$

Note: Adding rounding operations to this calculation process can further increase the accuracy.

3 Crystal temperature characteristic curve

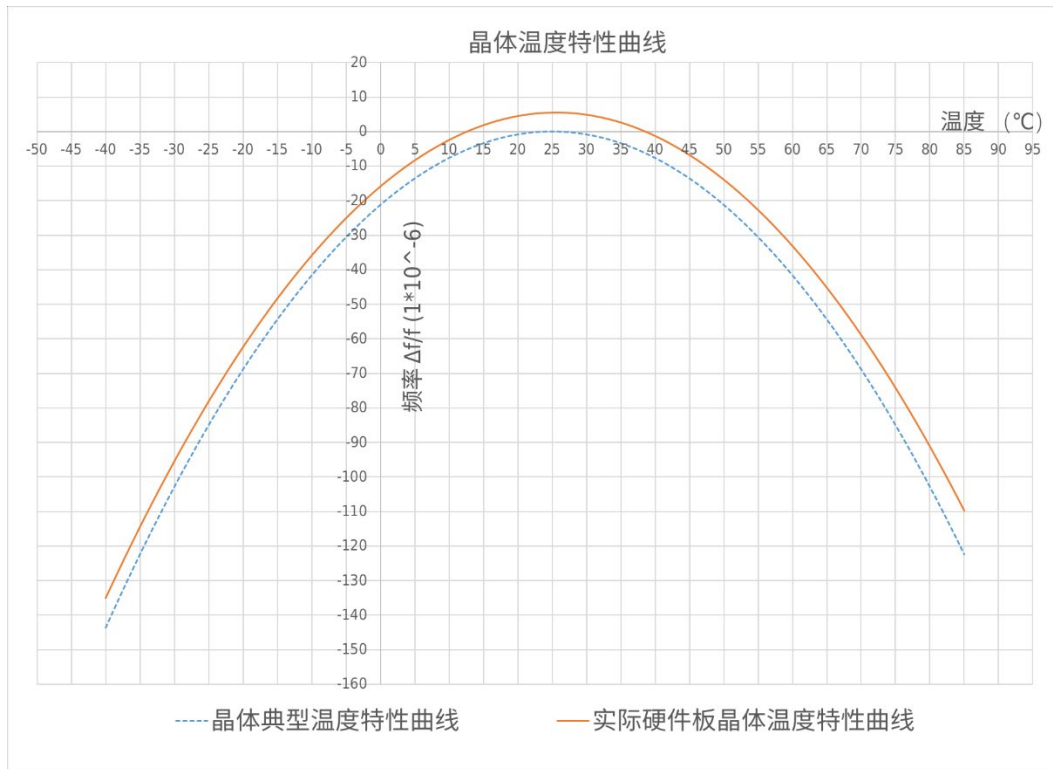


Figure 3-1 Crystal temperature characteristic curve

3.1 Typical crystal temperature characteristic curve

Applications with real-time clocks usually have an external 32768Hz high-precision ppm quartz crystal on the hardware board. Its temperature characteristic curve within a certain temperature range is approximately a parabola that opens downward. The more the temperature deviates from the vertex temperature, the greater the accuracy deviation of the crystal, as shown in the typical crystal temperature curve in the figure.

The typical crystal temperature curve can be expressed by the formula:

$$\Delta f / f = \alpha * (T - T_i)^2$$

Formula 4 Typical crystal temperature characteristic curve

- α : parabola coefficient, constant
- T: actual temperature (unit: °C)
- T_i : inflection point temperature (unit: °C), constant

According to the crystal specification, the values of α and T_i can be obtained. For example, for a certain crystal:

- $\alpha = -(0.034 \pm 0.0015)$, unit $10^{-6} / ^\circ\text{C}^2$
- $T_i = +25 \pm 5$, unit °C

Quartz crystals with good production consistency can use the same parameter to calculate the crystal accuracy deviation in application.

3.2 Actual hardware board crystal temperature characteristic curve

However, in actual applications, due to factors such as parasitic capacitance on the board and MCU pin capacitance, the actual temperature characteristic curve of the crystal will be offset, as shown in the actual hardware board crystal temperature characteristic curve in the figure.

The actual hardware board crystal temperature characteristic curve can be expressed by the formula:

$$\Delta f / f = \alpha * (T - T_i)^2 + ppm_t$$

Formula 5 Actual hardware board crystal temperature characteristic curve

- α : parabola coefficient, constant
- T: actual temperature (unit: °C)
- T_i : inflection point temperature (unit: °C), constant
- ppm_t: vertex PPM deviation, constant

In order to achieve the high precision requirements of the real-time clock in the entire operating temperature range, the three parameters α , T_i and ppm_t need to be obtained based on actual system measurements.

4 Real-time clock compensation method

This chapter introduces how to measure the crystal temperature characteristic curve in the application, calculate the ppm deviation of the crystal according to the current temperature, and compensate through the RTC module inside the microprocessor to achieve the correction of the actual output ppm.

The following is the process of real-time clock compensation. When used for the first time, it is necessary to measure the accuracy of the actual hardware board crystal at different temperatures, and fit the accuracy data with the temperature to generate a formula. The real-time compensation function is executed by the program. The software calculates the ppm deviation based on the current temperature and the crystal temperature characteristic formula, and then compensates the output accuracy through the RTC module inside the microprocessor.

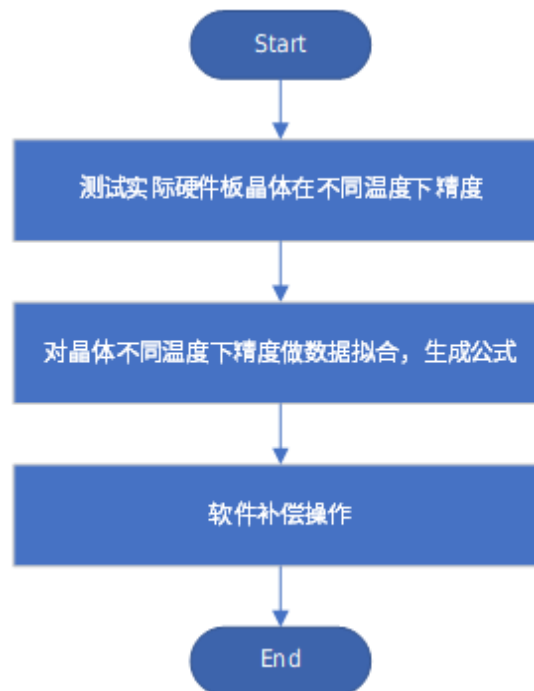


Figure 4-1 Real-time clock compensation process

4.1 Obtaining crystal temperature characteristic curve

In order to achieve the high-precision requirements of the real-time clock in the entire operating temperature range, the application needs to measure the temperature characteristic curve of the crystal, fit the formula and confirm the three parameters of α , T_i and ppm_t .

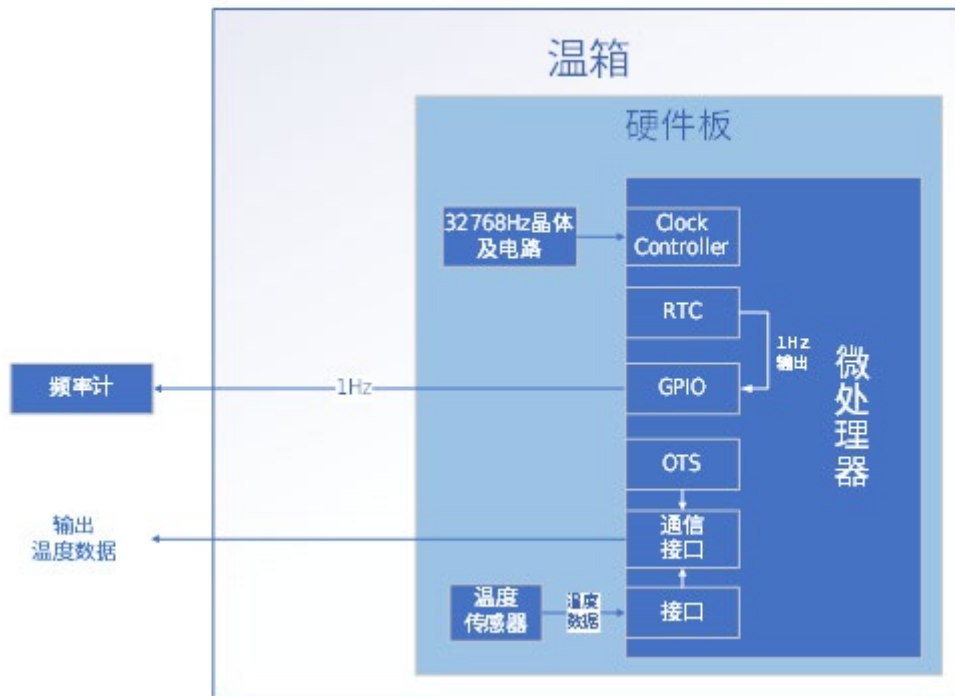


Figure 4-2 Crystal temperature characteristic curve test system block diagram

Note: Temperature data can be obtained from the on-chip OTS (On-chip Temperature Sensor) or external temperature sensor, which can be printed out through the communication interface or directly read in the debugger.

1) Test the accuracy of the actual hardware board crystal at different temperatures

Put the hardware board in the incubator, connect the frequency meter to 1Hz output, adjust the incubator at different temperatures and record the current temperature value and the ppm value of the frequency meter.

Notes:

- Because the crystal temperature characteristics determine that the more at both ends of the parabola, the greater the deviation of the crystal frequency changes with temperature. Therefore, under the requirement of high crystal accuracy, it is particularly important to pay attention to the accuracy of the recorded temperature here
- Place the temperature sensor as close to the crystal as possible

2) Fit the data for the accuracy of the crystal at different temperatures and generate a formula

Fit the crystal temperature characteristic formula based on the data recorded in the previous step.

Note: When the same formula cannot be used, different formulas can be used for different temperature zones.

As an example, the following unified formula will be used, such as

$$\Delta f / f = -0.0327 * (T - 25.6) ^ 2 + 5.6$$

Formula 6 Crystal temperature characteristic formula after measured data fitting

- $\alpha = -0.0327$
- $T_i = 25.6$
- $\text{ppm}_t = 5.6$

4.2 Software compensation operation

As shown in Figure 4-2, the crystal temperature characteristic curve test system block diagram, the microprocessor needs to obtain the current temperature, which can be obtained through the on-chip OTS or the off-chip temperature sensor.

The software calculates the crystal accuracy deviation at the current temperature according to the actual crystal temperature characteristic formula based on the current temperature value obtained, and then compensates the calculated crystal accuracy deviation value to the RTC compensation register. The process is as follows:



Figure 4-3 Software compensation process

- 1) Substitute the current temperature value into the actual crystal temperature characteristic formula $\Delta f / f = -0.0327 * (T - 25.6) ^ 2 + 5.6$, and calculate the ppm deviation of the crystal at the current temperature
- 2) Convert the ppm deviation value into a data value in units of RTC register compensation accuracy

The compensation referred to by RCT register 1 is $1 / 32768 / 2^5 = 0.954\text{ppm}$. Convert the result of 1) into the ppm deviation based on the RTC register through the calculation formula

- 3) Write the ppm deviation data based on the RTC register into the compensation value of the RTC compensation register routine:

```
/* RTC register macro definition */  
// 32768/32 --> 2^20  
#define RTC_PRECISION_BITS    20  
// 0 ppm value for RTC compensate register  
#define RTC_ZERO_VALUE        0x20  
// RTC compensate register bits = 9  
#define RTC_COMPEN_REG_BITS   9  
// Minimum RTC compensate value
```

```
#define RTC_COMPEN_MIN_VALUE  (uint16_t)((1 << (RTC_COMPEN_REG_BITS))
- 1)
// RTC shift bits = 7
#define RTC_COMPEN_SHIFT_BITS (16 - RTC_COMPEN_REG_BITS)
// Round up value
#define ROUND_UP_RTC_COMPEN_SHIFT (1 << (RTC_COMPEN_SHIFT_BITS - 1))

int32_t i32RtcCompenRValue;

uint16_t u16RtcRegRValue;

1)
// Calculate the crystal ppm deviation value  $\Delta f / f = -0.0327 * (T - 25.6) ^ 2 + 5.6$ 
2)
// Convert the ppm deviation value to a data value in units of RTC register
//compensation accuracy
// i32RtcCompenRValue * (2 ^ RTC_PRECISION_BITS) / 1000000
3)
// RTC compensation register compensation value calculation

u16RtcRegRValue    = (uint16_t)((int16_t)(RTC_ZERO_VALUE << RTC_COMPEN_SHIFT_BITS)
+ (int16_t)(i32RtcCompenRValue << RTC_COMPEN_SHIFT_BITS));

u16RtcRegRValue    = (u16RtcRegRValue + ROUND_UP_RTC_COMPEN_SHIFT >> RTC_COMPEN_SHIFT_BITS);
RTC->COMPEN_f.CR    = u16RtcRegRValue & RTC_COMPEN_MIN_VALUE;
```

Notes:

- The above routine is based on the register configuration of HC32L196 and contains schematic code
- The above sample is based on RTC->COMPEN_f.EN is set to 1, that is, the clock error compensation is enabled

So far, according to the pre-fitted crystal temperature characteristic formula, the microprocessor will obtain the current temperature into the formula to calculate the ppm deviation to be compensated, and then write the deviation value into the compensation value of the RTC register, thereby realizing the ppm compensation in the full operating temperature range.

5 Summary

This application note introduces the RTC compensation function and real-time clock compensation method of Xiaohua microprocessor to achieve the application requirements for high-precision real-time clock.

Version revision history

Revision	Revision Date	Revision content
Rev1.00	2024/12/11	First edition released.
Rev1.01	2025/04/02	In the applicable object section, HC32A448, HC32A472, HC32A460, and HC32A4A0 series models are added, and the product model description is modified.
Rev1.02	2025/04/11	HC32M441 series is added to the applicable objects.
Rev1.03	2025/05/30	1) Corrected the data error in section 2.2. 2) HC32L021 is added to the applicable objects.
Rev1.04	2025/06/04	HC32A052 series models are added to the applicable objects.